

static fields of research have room for creativity and innovation, and it provides a refreshingly new perspective on what may seem like a narrow field. Second, because it was designed and published anonymously and probably by someone outside the mainstream of research and practice, it reminds us that great thoughts and geniuses are found not only in fancy universities, established journals, and top firms, but also among the uncredentialed and unrecognized. Third, it represents a fascinating new generation of algorithms that are “native to computers,” meaning that they are not a translation of something that can be done with a cabinet and two hands like many old algorithms, but are fundamentally based on capabilities that are unique to computers (in this case, sleeping and threading). Fourth, the computer-native ideas it relies on (sleeping and threading) are very useful and worth putting in any algorithmicist’s toolbox for use in designing other algorithms. And fifth, I have an idiosyncratic affection for it, maybe just because it is a strange, creative misfit or maybe because I like its method of self-organizing order and the fact that I can use it if I’m ever in charge of saving a sinking ship.

From Sorting to Searching

Searching, like sorting, is fundamental to a variety of tasks in computer science (and in the rest of life). We may want to search for a name in a phone book, or (since we’re living after the year 2000) we may need to access a database and find a relevant record.

Searching is often merely a corollary of sorting. In other words, once we have sorted a list, searching is very straightforward—the sorting is often the hard part.

Binary Search

Binary search is a quick and effective method for searching for an element in a sorted list. It works a little like a guessing game. Suppose that someone is thinking of a number from 1 to 100 and you are trying to guess it. You may guess 50 as your first guess. Your friend says that 50 is incorrect but allows you to guess again and gives you a hint: 50 is too high. Since 50 is too high, you guess 49. Again, you are incorrect, and your friend tells you